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Discussion Questions:

1. Why does the current implementation use StackFrontier instead of QueueFrontier?

In this line (127) we can choose either StackFrontier() or QueueFrontier() – Now it's StackFrontier() which is DFS, and we use it because it uses less memory.

2. What would happen if you changed line 127 to use QueueFrontier()?

It uses BFS (Queue) instead of DFS (Stack).

The search behavior changes because BFS explores nodes level by level, spreading out across all nearby options first, rather than going deep into one path like DFS.

The resulting path may not be the same. However, BFS has the advantage of always finding the shortest path (in terms of number of steps), while DFS does not guarantee that.

- Would it explore more or fewer states?

It depends on the maze, but till now it explored fewer states

3. Is DFS guaranteed to find the shortest path? Why or why not?

DFS(Stack) not guaranteed to find the shortest path and may explore very long path, because it goes deeply.

4. How does the number of explored states differ between maze1.txt and maze2.txt?

- Run both and compare the "States Explored" output

1	11	11
2	194	77

- Why is there a difference?

Because each one has different maze structures and have different goal.

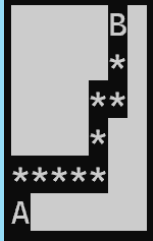
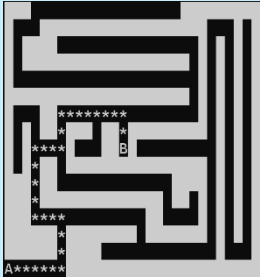
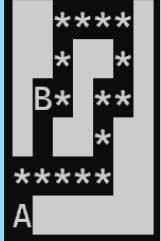
5. What would happen if we removed the explored set check?

- Hint: Think about cycles in the maze

In some cases it might go infinitely (infinite loop), because it may visit the same cell many times without noticing that it was visited before.

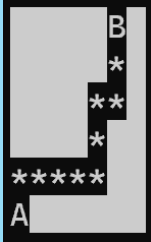
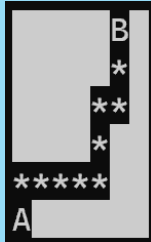
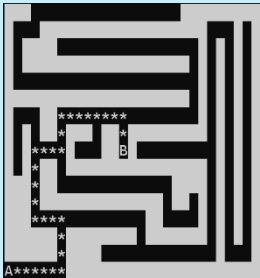
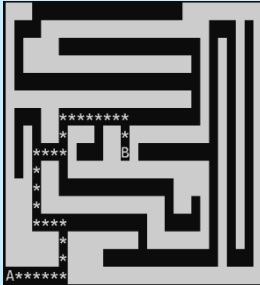
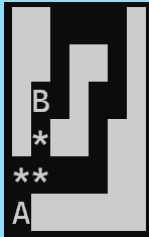
Exercise 1: Run and Observe

1. Run the program with all three maze files
2. Record the number of states explored for each
3. Compare the solution paths in the generated images

Maze \ Algorithm	DFS
1	 States Explored: 11
2	 States Explored: 194
3	 States Explored: 17

Exercise 2: Switch to BFS

1. Modify line 127 to use `QueueFrontier()` instead of `StackFrontier()`
2. Run the program again with all maze files
3. Compare: states explored, path length, and path shape

Maze \ Algorithm	DFS	BFS
1	 States Explored: 11	 States Explored: 11
2	 States Explored: 194	 States Explored: 77
3	 States Explored: 17	 States Explored: 6

4. Document your findings

While both algorithms use different ways of searching, but I noticed that in maze 1 and 2 both DFS and BFS have the same result path regardless of the number of States Explored.

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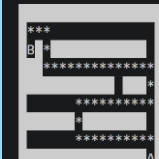
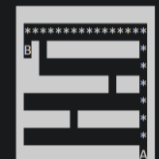
Exercise 3: Create Your Own Maze

1. Create a new file maze4.txt with your own maze design
2. Make sure it has exactly one A and one B
3. Test it with both DFS and BFS
4. Design a maze where DFS and BFS find different paths

{ Maze4.txt }:

```

1 #####
2 #
3 #B# ##### #
4 ###
5 ##### ## #
6 #
7 ##### #
8 #
9 #####A#
10
    
```

Maze \ Algorithm	DFS	BFS
4	States Explored: 61 Solution: 	States Explored: 70 Solution: 

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Lab 1

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